Supplementary Material: Benchmark Definitions, Execution Protocol, and Uncertainty

The Grid-GDP functions are adapted from the Quantum Hamiltonian Descent collection; each instance is defined by the expression and domain implemented there, which can differ from other functions known by the same name (SECQUOIA, 2026a; Wu Group, n.d.).

S1 Coordinate domains and normalization

The coordinate mapping and objective normalization follow Section 3.4 of the main paper. Both coordinate endpoints are included, and L and U are the minimum and maximum over the 961 lattice points. All trigonometric arguments are in radians. Full-precision normalization constants and target coordinates are archived in `grid_inventory.json` (Section S8).

Table S1: Grid-GDP: coordinate domains and target-set sizes of the 21 test functions of Section S2. Each coordinate range is sampled at 31 equally spaced points that include both endpoints, which gives a 31×31 lattice of 961 configurations and 62 Boolean indicators. The target set of a function contains the lattice points whose normalized objective $F = (f - L)/(U - L)$ is at most 10^{-5}, and a run reaches the reference target of Equation (8) of the main paper when it evaluates one of these points.

Function	x domain	y domain	n_{target}
Ackley	$[-5, 5]$	$[-5, 5]$	1
Alpine 1	$[0, 10]$	$[0, 10]$	1
Alpine 2	$[0, 1000]$	$[0, 1000]$	1
Bohachevsky 2	$[-100, 100]$	$[-100, 100]$	1
Three-hump camel	$[-5, 5]$	$[-5, 5]$	1
Csendes (squared variant)	$[-1, 1]$	$[-1, 1]$	36
Deflected corrugated spring	$[0, 10]$	$[0, 10]$	1
Drop-Wave	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Easom	$[-10, 10]$	$[-10, 10]$	1
Griewank	$[-600, 600]$	$[-600, 600]$	1
Hölder table	$[-10, 10]$	$[-10, 10]$	4
Hosaki	$[0, 5]$	$[0, 6]$	1
Lévy	$[-10, 10]$	$[-10, 10]$	2
Lévy 13	$[-10, 10]$	$[-10, 10]$	4
Michalewicz	$[0, \pi]$	$[0, \pi]$	1
Rastrigin	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Rosenbrock	$[-2, 2]$	$[-1, 3]$	2
Shubert (variant)	$[0, 10]$	$[0, 10]$	1
Styblinski–Tang	$[-5, 5]$	$[-5, 5]$	1
Sum of squares	$[-10, 10]$	$[-10, 10]$	1
Xin-She Yang 3 (variant)	$[-20, 20]$	$[-20, 20]$	870

x domain, y domain: coordinate ranges; n_{target}: number of lattice points in the target set; F: normalized objective; L, U: minimum and maximum of the objective f over the lattice.

S2 Grid-GDP objective functions

The implemented functions below use the domains in Table S1 (SECQUOIA, 2026a).

1. Ackley

$$f(x,y) = -20\exp\left(-0.2\sqrt{\frac{x^2+y^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x)+\cos(2\pi y)}{2}\right) + 20 + e. \tag{S1}$$

2. Alpine 1

$$f(x,y) = |x\sin x + 0.1x| + |y\sin y + 0.1y|. \tag{S2}$$

3. Alpine 2

$$f(x,y) = -\sqrt{xy}\,\sin x \sin y. \tag{S3}$$

4. Bohachevsky 2

$$f(x,y) = x^2 + 2y^2 - 0.3\cos(3\pi x)\cos(4\pi y) + 0.3. \tag{S4}$$

5. Three-hump camel

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2. \tag{S5}$$

6. Csendes (squared variant)

$$f(x,y) = c(x) + c(y), \qquad c(t) = \begin{cases} t^6\left[2+\sin(1/t)\right]^2, & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{S6}$$

The GDP uses branch-fixed auxiliaries $r_t = 1/t$ for $t \neq 0$ and $r_t = 0$ otherwise, avoiding division by zero without an epsilon shift.

7. Deflected corrugated spring

With $r = \sqrt{(x-5)^2 + (y-5)^2}$,

$$f(x,y) = 0.1r^2 - \cos(5r). \tag{S7}$$

8. Drop-Wave

$$f(x,y) = -\frac{1+\cos\left(12\sqrt{x^2+y^2}\right)}{0.5(x^2+y^2)+2}. \tag{S8}$$

9. Easom

$$f(x,y) = -\cos x \cos y\, \exp\left(-[(x-\pi)^2 + (y-\pi)^2]\right). \tag{S9}$$

10. Griewank

$$f(x,y) = \frac{x^2+y^2}{4000} - \cos x \cos\left(\frac{y}{\sqrt{2}}\right) + 1. \tag{S10}$$

11. Hölder table

$$f(x,y) = -\left|\sin x \cos y\, \exp\left(\left|1 - \frac{\sqrt{x^2+y^2}}{\pi}\right|\right)\right|. \tag{S11}$$

12. Hosaki

$$f(x,y) = \left(1 - 8x + 7x^2 - \frac{7}{3}x^3 + \frac{1}{4}x^4\right) y^2 \exp(-y). \tag{S12}$$

13. Lévy

Define $w_x = 1 + (x-1)/4$ and $w_y = 1 + (y-1)/4$. Then

$$\begin{aligned} f(x,y) = {} & \sin^2(\pi w_x) \\ & + (w_x - 1)^2\big[1 + 10\sin^2(\pi w_x + 1)\big] \\ & + (w_y - 1)^2\big[1 + \sin^2(2\pi w_y)\big]. \end{aligned} \tag{S13}$$

14. Lévy 13

$$\begin{aligned} f(x,y) = {} & \sin^2(3\pi x) + (x-1)^2\big[1 + \sin^2(3\pi y)\big] \\ & + (y-1)^2\big[1 + \sin^2(2\pi y)\big]. \end{aligned} \tag{S14}$$

15. Michalewicz

$$f(x,y) = -\sin x\, \sin^{20}\left(\frac{x^2}{\pi}\right) - \sin y\, \sin^{20}\left(\frac{2y^2}{\pi}\right). \tag{S15}$$

16. Rastrigin

$$f(x,y) = x^2 - 10\cos(2\pi x) + y^2 - 10\cos(2\pi y) + 20. \tag{S16}$$

17. Rosenbrock

$$f(x,y) = (1-x)^2 + 100(y - x^2)^2. \tag{S17}$$

18. Shubert (variant)

$$\begin{aligned} f(x,y) = {} & \big[\cos(2x+1) + 2\cos(3x+2) + 3\cos(4x+3)\big] \\ & \times \big[\cos(2y+1) + \cos(y+2)\big]. \end{aligned} \tag{S18}$$

19. Styblinski–Tang

$$f(x,y) = \frac{1}{78}\,\frac{1}{2}\left(x^4 - 16x^2 + 5x + y^4 - 16y^2 + 5y\right). \tag{S19}$$

20. Sum of squares

$$f(x,y) = x^2 + 2y^2. \tag{S20}$$

21. Xin-She Yang 3 (variant)

$$f(x,y) = \exp\left(-\left[\left(\frac{x}{15}\right)^6 + y^6\right]\right) + 2\exp\big(-(x^2+y^2)\big)\cos^2 x \cos^2 y. \tag{S21}$$

S3 Executable protocol and implementation settings

Python and NumPy use the job seed, and RBFOpt also receives rand_seed. BAM exposes no separate native random-seed option: its process wrapper seeds Python/NumPy, while the Grid seed identifies the common initial design. FMQA-SA uses three NumPy SeedSequence(seed) streams for FM initialization, annealing, and uniform unseen-point fallback; each training or annealing invocation draws a seed from its stream. PyTorch uses deterministic CPU operations and one thread.

Grid D-SDA enumerates L_∞ neighbors lexicographically from $(-1, 0, 1)$, excluding the zero direction and out-of-bounds points; ties select the first point.

The RBFOpt options are rbf=auto, global_search_method=genetic, do_init_strategy=False, ga_base_population_size=400, and ga_num_generations=20.

BAM receives x0=None and max_evals=B-3; its local models use ALAMO/BARON, whose internal versions are not reported by the package.

The FMQA-SA time limit is checked between operations and through training/sampling callbacks. Each FM uses identity activation, zero bias, and normal initial weights with mean zero and standard deviation 0.01. Training uses Adam (Kingma and Ba, 2015), 1,000 full-batch epochs, mean squared error, and learning rate 0.01. Coordinate ξ_j uses $\lfloor \log_2 u_j \rfloor + 1$ unsigned bits, where $u_j > 0$ is its upper bound; QUBO coefficients are divided by the largest absolute entry of the quadratic matrix. D-Wave's SimulatedAnnealingSampler in dwave-samplers 1.6.0 uses 100 reads, 1,000 sweeps, and a geometric inverse-temperature schedule (D-Wave Systems, n.d.). All process workers and the Grid FMQA-SA worker have a 2,100 s watchdog. The Grid RBFOpt/BAM trace watchdogs are disabled (max_run_seconds=0); archived Grid D-SDA uses budget, target, or local-search termination without an additional method-clock limit.

S3.1 Versions and independently timed process evaluations

Software versions are given in Section 3.6 of the main paper.

The conventional GDP process models retain their archived initializations, including constructor defaults for CSTR30; the updated DFO evaluation table is identified separately in Section S8.

The acceptance rule in gdp_models/evaluation.py requires all of the following: a finite objective; termination in {optimal, locallyOptimal, globallyOptimal}; solver status other than error or aborted; interface status SOLVESTAT=1 and MODELSTAT$\in \{1, 2\}$; and a finite maximum primal violation no greater than 10^{-5}. Time-limit, iteration-limit, evaluation-limit, and error outcomes are not accepted, even if a finite feasible candidate is available.

The primal check uses unscaled violations of original constraints and initialized-variable bounds, including constraint rows deactivated during preprocessing. Nonfinite objectives, constraint bodies, or initialized-variable values preclude acceptance; unused uninitialized variables are skipped. Preprocessing rejection or worker failure supplies no accepted objective.

The CSTR initializer cstr_box_ipopt.py:initialize_feasible runs after preprocessing and checks the mass-balance/zero-recycle start at 10^{-8} without changing constraints or bounds. All process evaluations use cold starts; NLP solutions are not transferred between configurations. Objectives, acceptance status, and NLP point costs are linked to the same recorded attempt by model, CSTR size, and coordinate, without successful-retry selection or cross-size time substitution.

S3.2 Observed proposal and initialization behavior

Table S2 pools FMQA-SA proposals across budgets and seeds, excluding the shared initialization and including terminal callbacks. Fallback supplies 52.89% of process proposals; the uniform comparison evaluates the complete procedure, without isolating the contributions of learning, annealing, and fallback.

Table S2: Grid-GDP and process GDP models: proposals and uniform fallback proposals of `FMQA-SA`, pooled over all budgets and seeds, with 1,050 Grid-GDP runs (21 functions, five budgets, and ten seeds) and ten runs per process condition. A proposal is a configuration that `FMQA-SA` selects after the three shared initial configurations, so a run with evaluation budget B makes $B - 3$ proposals. A fallback is a proposal drawn uniformly from the unevaluated configurations because no annealed candidate is a new configuration within the coordinate bounds; the rate is the percentage of proposals that are fallbacks, and the process total pools the four process models.

Benchmark	Runs	Proposals	Fallbacks	Rate (%)
Grid-GDP	1,050	51,450	9,522	18.51
`CSTR`	60	1,350	369	27.33
`Column`	10	100	86	86.00
`Small Batch`	10	70	58	82.86
`Dynamic`	60	1,460	1,063	72.81
Process total	140	2,980	1,576	52.89

Runs: number of `FMQA-SA` runs.

S4 Process model scale, units, and discrete decisions

Table S3 counts the original GDP before reformulation or presolve, including all active disjunct branches and fixed continuous variables. `Dynamic` is discretized with 30 finite elements per stage and three Radau points. Binary counts refer to the indicator variables already constructed by the model, before Boolean-to-binary conversion. All nine models have zero additional nonbinary integer variables.

Table S3: Process GDP models: numbers of variables, constraints, and disjunctions of the original GDP models. The counts are taken before reformulation or presolve and include all active disjunct branches and fixed continuous variables, and `Dynamic` is discretized with 30 finite elements per stage and three Radau collocation points.

Model	N	n_c	n_b	m_a	m_l	n_d
`CSTR`	5	56	20	100	12	10
	10	106	40	195	22	20
	15	156	60	290	32	30
	20	206	80	385	42	40
	25	256	100	480	52	50
	30	306	120	575	62	60
`Column`	–	446	30	637	16	15
`Small Batch`	–	19	18	34	3	9
`Dynamic`	–	1,647	27	3,276	31	9

N: number of potential reactors of `CSTR`; –: not applicable. n_c: continuous variables; n_b: binary indicator variables constructed by the model; m_a: algebraic constraints; m_l: logical constraints; n_d: disjunctions.

The `Column` objective is $10^3(Q_c + Q_b) + 10^3 n_{\text{active}}$, the weighted cost proxy of the source model, which we did not convert to an annualized cost. The `Dynamic` objective is $-\int_0^9 x(t)^2\,dt$.

S5 Lomustine simulation-based benchmark

The `Lomustine` configurations and responses come from the corrected-model tables of SECQUOIA (2026b), generated with `PharmaPy` at commit 3194efd. For each superstructure option, the source ran `PyNomadBBO` from a stored initial point on the normalized operating inputs with 60 simulations and stored the best objective, the constraint values, the equipment sizes, and the wall time of the inner optimization. The inner search accepts a point when the solubility path constraint holds. The heptane mass-fraction specification

of the source is not enforced, because the fixed three-hour vaporizer operation cannot reach it within the pressure bounds. Table S7 lists the 36 two-reactor configurations; the four single-reactor semibatch options of the source do not fit the three external variables and are excluded. The configurations with a batch crystallizer have the lowest objective for every reactor pair, and the choice of the first reactor changes the objective by up to a factor of 2.9. The vendored table, case module, run records, and analysis are listed in Section S8.

Option order and objective. The external variables index categorical alternatives, so we repeated the 210 runs with the five other orders of Table S4, using the same seeds, initial designs in index space, and budgets. Table S5 reports the hits in the source order and their range over the six orders. `NOMAD` ended with a segmentation fault after identical evaluations on both attempts in 12 runs over all campaigns; as for the Grid runs of Section S12, these runs are scored on the evaluations completed before the crash. Table S6 repeats the source-order runs with the bare-module capital cost of each flowsheet at its optimized operation as the objective, whose reference is the all-batch flowsheet at \$82,766. This cost is evaluated at the operating point that maximizes production and crystal size, and it is not itself optimized over the operating inputs.

Table S4: `Lomustine`: the six orders of the reactor and crystallizer alternatives used as external-variable indices. An order assigns consecutive index values to the alternatives of each external variable in the listed sequence. The source order is used in the main comparison, whose runs are repeated in the five other orders with the same seeds, initial designs in index space, and budgets.

Order	Reactors	Crystallizers
Source	batch, CSTR, PFR	batch, 1, 2, 3
Reversed	PFR, CSTR, batch	3, 2, 1, batch
Back-mixing	batch, PFR, CSTR	batch, 3, 2, 1
Random 1	PFR, batch, CSTR	1, 2, batch, 3
Random 2	batch, PFR, CSTR	batch, 3, 1, 2
Random 3	CSTR, PFR, batch	2, 1, batch, 3

Source: order of the source enumeration; Reversed: source order backward; Back-mixing: alternatives ordered by degree of back-mixing; Random 1 to 3: three further permutations. CSTR: continuous stirred-tank reactor; PFR: plug-flow reactor; 1, 2, 3: number of mixed-suspension, mixed-product-removal (MSMPR) crystallizer stages.

Table S5: `Lomustine`: target hits of the six DFO methods and uniform sampling in the source order and over the six option orders of Table S4, with ten runs per method, budget, and order. A target hit is a run that reaches the reference target of Equation (8) of the main paper, here the reference configuration of Table S7. All orders share the seeds, initial designs in index space, and budgets of the source order, and the rows for the fewest and most target hits give the smallest and largest count over the six orders.

		Target hits (of 10)						
B		`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform
10	Source order	1	6	8	7	0	2	2
	Fewest over orders	1	5	2	4	0	2	0
	Most over orders	6	6	8	7	2	7	4
15	Source order	4	6	9	10	4	3	4
	Fewest over orders	3	5	7	5	0	3	2
	Most over orders	8	9	10	10	10	10	4
20	Source order	4	6	10	10	6	7	5
	Fewest over orders	3	5	9	6	0	7	4
	Most over orders	8	9	10	10	10	10	7

B: evaluation budget; Uniform: uniform sampling without replacement; Source order: order of the source enumeration, used in the main comparison.

Table S6: `Lomustine` with the capital-cost objective: target hits of the six DFO methods and uniform sampling in the source order, with ten runs per method and budget. The capital-cost objective is the bare-module capital cost of each flowsheet at its optimized operation, and a target hit is a run that reaches the reference of this objective, the all-batch configuration at \$82,766.

	Target hits (of 10)						
B	`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform
10	3	9	7	8	**10**	2	4
15	6	9	8	**10**	**10**	3	7
20	9	9	**10**	**10**	**10**	5	8

B: evaluation budget; Uniform: uniform sampling without replacement. Bold: most target hits for each budget, including ties.

Table S7: `Lomustine`: objective of every configuration after the simulation-based optimization of its operating inputs, and the wall time of that optimization, with one optimization per configuration. The objective is $-(\dot{m}/10^{-4} + 2d_{50})$, and each configuration combines a first reactor, a second reactor, and a crystallizer. The reference configuration, a plug-flow reactor followed by a batch reactor and the batch crystallizer, has the lowest objective.

		Objective				Wall time (s)			
R01	R02	batch	1 MSMPR	2 MSMPR	3 MSMPR	batch	1 MSMPR	2 MSMPR	3 MSMPR
	batch	-170.68	-131.70	-146.86	-148.92	91	310	538	835
batch	CSTR	-230.52	-175.25	-175.27	-166.13	146	348	650	970
	PFR	-268.46	-209.36	-213.29	-202.26	159	348	651	975
	batch	-483.85	-360.16	-365.91	-345.28	152	331	656	936
CSTR	CSTR	-331.90	-244.39	-245.34	-239.50	159	367	660	1,019
	PFR	-429.81	-315.22	-321.46	-304.62	206	408	717	1,017
	batch	**-487.84**	-362.45	-361.55	-348.06	208	347	655	886
PFR	CSTR	-351.71	-249.82	-259.47	-244.65	183	388	659	983
	PFR	-433.41	-318.34	-324.18	-306.21	211	393	629	907

$\dot{m}$: production rate (kg/s); d_{50}: median crystal size (μm); R01, R02: first and second reactor; CSTR: continuous stirred-tank reactor; PFR: plug-flow reactor; batch, 1 to 3 MSMPR: batch crystallizer or one to three mixed-suspension, mixed-product-removal (MSMPR) stages. Bold: reference configuration.

S6 Catalytic distillation column

Corrected model. We checked every equation block, parameter table, and bound of the published model against standard forms and literature values, and the `Catalytic Column` benchmark uses the model with the errors fixed. The corrections concern the vapor–liquid equilibrium and the equilibrium constant of the reaction, the mixture liquid density (Rackett equation with the Spencer–Danner mixing rule) and surface tension (Macleod–Sugden), the tray hydraulics, and the objective, which counted the annualization factor twice and credited the impurities in the bottoms at the ETBE price. The pure-component data of isobutene, 1-butene, and ethanol are fitted to open reference equations of state, and the ETBE data, for which no reference equation exists, agree with open measurements of its vapor pressure and density. The bottoms must contain at least 95 mol% ETBE, the fuel-grade specification, the catalytic stages stay at or below 120 °C, the maximum operating temperature of the Amberlyst 15 catalyst, and the prices are 2010 market prices. The repository lists each finding with its effect on the published optimum; with the 95 mol% specification, the published optimal configuration is infeasible, because its column is too short.

Evaluation campaign. The 210 runs made 12,015 evaluations: 6,921 returned accepted objectives, 2,789 were logical rejections without an NLP solve, and 2,305 returned unaccepted NLP responses. Eight `D-SDA` runs stopped before their budgets because no neighbor improved, and all other runs used their full budgets. The runs requested 3,499 distinct configurations, of which 2,652 were accepted, and solving them took 7.7 h. The completion times of the runs sum to 20.2 h, but the campaign took 2.8 h on three cores because a configuration requested by several runs was solved once. Each run evaluated distinct configurations, so the reuse comes from different runs requesting the same configuration.

Search box. The GDP admits 100,947 configurations, and the gap box $g_i \leq 6$ of the search methods keeps 54,159 of them, including every design within 5% of the best of a pilot of 202 configurations. The reference has $g_1 = f_1 - 2 = 6$, on the bound of the box, so a better design could lie just outside it. We therefore solved, with the initialization and acceptance rule of the campaign, all 747 configurations outside the box within two stages of the reference in every stage position. Of these, 255 are accepted and 492 are locally infeasible, and none is better than the reference: the best, $(9, 9, 13, 17, 17, 21)$ at 24,930.96 \$/yr, lies 0.14% above it. This check ran on the server of the campaign with its `GAMS` and `CONOPT` versions, and the longest solve took 40 s, below the campaign's 300 s limit; an earlier run of the same check on a workstation with `GAMS` 54.2.2 returned the same status for every configuration and the same objective for every accepted one.

MINLP forms. The published MINLP form makes the stage decisions binary variables, multiplies switched equations by these variables or by the stage-existence indicator, and keeps the property and kinetic equations on every stage. The disjunctive form behind the Big-M and Hull reformulations has five disjunctions for each internal stage, one per decision, whose terms collect the equations that the GDP model switches on or off (Table S8). The property and feed-enthalpy equations hold on every stage, as in the published form, but the kinetics stay in the catalyst disjunction; with the kinetics on every stage, the bounds of the reaction rate restricted the temperature of a stage without catalyst, and with the binary variables of $(3, 4, 5, 11, 11, 14)$ fixed, the Big-M reformulation stopped at 32,075 \$/yr, a point from which the GDP form continues to 30,822 \$/yr. Each nonlinear disjunct equation $h(\mathbf{x}) = 0$ or $h(\mathbf{x}) \leq 0$ is replaced by the definition $s\,r = h(\mathbf{x})$, which holds on every stage, and by the linear disjunct constraint $r = 0$ or $r \leq 0$; a linear disjunct equation is divided by s. The scale s is 1.01 times the largest absolute value of h over the variable bounds, computed by interval arithmetic after feasibility-based bound tightening with the constraints that hold for every configuration. Every scaled disjunct constraint therefore lies in $[-1, 1]$ at every feasible point, so $M = 1$ is a valid Big-M value for all of them, and the bound tightening adds 101 implied variable bounds that are written into the model. The scales range from 0.10 to 1.3×10^{18}; the largest belongs to the rate equation, whose products of activities, adsorption constant, and rate span this range over the variable bounds. The interval bounds of the flooding and weeping correlations are finite only if the liquid is denser than the vapor, which the source bounds do not imply, so the disjunctive form bounds the liquid and vapor mass densities by $\rho_L \geq 200$ and $5 \leq \rho_V \leq 150$ kg/m^3; every validated solution has $\rho_L \geq 529$ and $\rho_V \leq 34$ kg/m^3. These bounds belong only to the Big-M and Hull reformulations; the GDP model, its fixed-configuration evaluations, and the published MINLP form do not impose them. Within

the model's bounds (pressure 9.5 to 10 bar, temperature 200 to 445 K, compressibility factor 0.5 to 1.3) and for any vapor composition, whose molar mass lies between 46.07 and 102.18 g/mol, the vapor density lies between 9 and 123 kg/m^3, so the vapor bounds cannot bind and only the liquid bound restricts the reformulations. The `GAMS` convex hull reformulation accepts only inequalities, so each disjunct equality is written as two inequalities, and Big-M of the pair equals Big-M of the equality. Table S9 compares the forms with the binary variables fixed.

Table S8: `Catalytic Column`: the five disjunctions of each internal stage k in the disjunctive form behind the Big-M and Hull reformulations. Each row is one disjunction, whose Boolean indicator enforces the equations of the second column on stage k when it is true and those of the third column when it is false. The property and feed-enthalpy equations hold on every stage and belong to no disjunction.

Indicator	Indicator true	Indicator false
stage k exists	vapor–liquid equilibrium, tray hydraulics, pressure drop, flooding and weeping limits, tray height	vapor composition and flow pass through, no pressure drop, no tray height
catalyst on stage k	kinetics, catalyst hydraulics and flooding, reaction terms, 120 °C limit	no reaction, no catalyst pressure drop
ethanol feed at k	feed flow	no feed flow
butenes feed at k	feed flow	no feed flow
boil-up return at k	boil-up flow	no boil-up flow

k: internal stage of the column; Indicator: Boolean indicator of the disjunction.

Table S9: `Catalytic Column`: objectives of four model forms with the binary variables fixed at nine configurations, with one `CONOPT` solve per form and configuration from the initialization of the published model. The four forms are the GDP model evaluated by the search methods, the MINLP form published with the original `D-SDA` study of this column, and the Big-M and Hull reformulations of the disjunctive form of Table S8. The first row is the reference configuration.

	Objective ($/yr)			
$(f_1, c_1, c_2, c_3, f_2, b)$	GDP	Published MINLP	Big-M	Hull
$(8, 8, 12, 17, 17, 21)$	24,895.55	24,895.55	24,895.55	24,895.55
$(7, 8, 11, 16, 16, 20)$	25,271.33	25,271.33	25,271.33	25,271.33
$(3, 4, 5, 11, 11, 14)$	30,822.02	30,822.02	30,822.02	30,822.02
$(4, 4, 9, 10, 14, 17)$	28,373.98	28,373.98	28,373.98	28,373.98
$(3, 6, 12, 14, 19, 21)$	31,116.60	32,320.55	31,116.60	31,116.60
$(2, 3, 9, 10, 20, 21)$	35,745.43	35,194.65	35,745.43	35,745.43
$(4, 4, 6, 8, 8, 10)$		infeasible		
$(3, 4, 5, 6, 8, 9)$		infeasible		
$(13, 13, 15, 16, 17, 19)$		infeasible		

f_1, f_2: stages of the ethanol and butenes feeds; $c_1 < c_2 < c_3$: catalytic stages; b: stage of the boil-up return. infeasible: every form reports the configuration as locally infeasible.

MINLP solvers. Table S10 lists the results of the MINLP solvers. `SBB` stops on the Big-M and Hull reformulations when the NLP relaxations of its nodes exceed the limit on function evaluations outside their domains. Table S11 lists the global optimality certificates that `SCIP` and `BARON` return when they start from a feasible design of the published form; the `GAMS Examiner` confirms that both starting designs satisfy every constraint and bound within 10^{-6}.

Table S10: `Catalytic Column`: outcomes and solve times of four MINLP solvers on three model forms, with one run per solver and form started from the initial configuration $(3, 4, 5, 6, 8, 9)$ of the published model. The outcome is the objective and configuration of the design found, or the reason no feasible design was returned; the reference design has an objective of 24,895.55 \$/yr. The time is the wall-clock time of the `GAMS` job, and each run uses one core and a 3,600 s time limit.

Form	Solver	Outcome	Time (s)
Published	`DICOPT`	no feasible design; every NLP subproblem infeasible, iteration limit	686
	`SBB`	28,717.22 \$/yr at $(4, 4, 5, 6, 6, 9)$	10
	`BARON`	declared infeasible	255
	`SCIP`	declared infeasible	9
Big-M	`DICOPT`	no feasible design; NLP subproblem failed	902
	`SBB`	no feasible design; domain violation limit	38
	`BARON`	declared infeasible	214
	`SCIP`	no feasible design; lower bound -5.7×10^7 \$/yr	3,600+
Hull	`DICOPT`	no feasible design; NLP subproblem failed	211
	`SBB`	no feasible design; domain violation limit	130
	`BARON`	declared infeasible	362
	`SCIP`	no feasible design; lower bound -5.7×10^7 \$/yr	3,600+

Published: published MINLP form; Big-M, Hull: reformulations of the disjunctive form; configurations $(f_1, c_1, c_2, c_3, f_2, b)$ as in Table S9; 3,600+: stopped at the time limit.

Table S11: `Catalytic Column`: global optimality certificates that `SCIP` and `BARON` return for the published MINLP form, with one run per solver from each of two feasible starting designs. The first starting design comes from the pilot of 202 configurations and the second is the reference configuration. The status and objective are those reported by each solver, and the time is its solve time.

	`SCIP`			`BARON`		
Starting design	Status	Objective (\$/yr)	Time (s)	Status	Objective (\$/yr)	Time (s)
$(7, 8, 11, 16, 16, 20)$	optimal	25,271.33	8	optimal	25,271.33	172
$(8, 8, 12, 17, 17, 21)$	optimal	24,895.55	10	optimal	24,895.55	270

Starting design: configuration $(f_1, c_1, c_2, c_3, f_2, b)$ as in Table S9.

S7 Completion costs

Table S12 reports the search times in all 14 process conditions; the completion times are in Table 2 and the enumeration times in Table 1 of the main paper. Table S13 reports the Grid-GDP costs at $B = 100$. `MISO` reaches the most Grid targets, but its median search time of 32.70 s is about four times that of `RBFOpt`, whereas `D-SDA` and `NOMAD` stop after medians of 37 and 31.5 evaluations and have the lowest median completion times, 18.38 and 18.50 s (`NOMAD` over its 189 timed runs).

Per-function Grid results at $B = 100$ and complete seed-level outcomes are available in the analysis files linked in Section S8.

Table S12: Process GDP models: search times of the six DFO methods and uniform sampling, with ten runs per method and condition. The search time of a run is the time the method spends selecting configurations, which excludes the cost of evaluating them, and the medians and percentiles are over the ten runs. Table 2 of the main paper gives the full-run completion times and target hits.

Model	N	B	Search time	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
CSTR	5	8	Median (s)	0.0105	0.0367	0.0596	0.659	0.0360	3.66	0.00460
			25th percentile (s)	0.0102	0.0323	0.0590	0.658	0.0151	3.64	0.00387
			75th percentile (s)	0.0111	1.13	0.0597	0.669	0.0816	3.67	0.00478
	10	15	Median (s)	0.0137	1.83	0.302	0.677	0.137	8.00	0.00807
			25th percentile (s)	0.0130	0.469	0.288	0.675	0.126	7.98	0.00772
			75th percentile (s)	0.0147	1.88	0.315	0.683	0.158	8.08	0.00812
	15	22	Median (s)	0.0169	0.825	1.00	3.29	0.150	12.4	0.0113
			25th percentile (s)	0.0156	0.0949	0.492	1.21	0.140	12.3	0.0108
			75th percentile (s)	0.0172	1.82	3.26	7.50	0.162	12.4	0.0116
	20	29	Median (s)	0.0207	1.63	2.82	10.8	0.167	16.9	0.0139
			25th percentile (s)	0.0189	0.521	0.632	9.02	0.165	16.8	0.0121
			75th percentile (s)	0.0218	1.92	4.01	15.8	0.313	16.9	0.0148
	25	36	Median (s)	0.0258	1.08	2.47	17.6	0.401	21.3	0.0161
			25th percentile (s)	0.0238	0.170	1.86	16.4	0.392	21.3	0.0147
			75th percentile (s)	0.0299	2.32	4.43	18.8	0.449	21.4	0.0186
	30	43	Median (s)	0.0585	1.90	4.99	18.1	0.425	25.8	0.0241
			25th percentile (s)	0.0556	1.76	2.95	14.0	0.404	25.7	0.0224
			75th percentile (s)	0.0610	7.65	10.3	19.3	0.451	26.0	0.0257
Column	–	13	Median (s)	0.0120	0.0423	0.226	0.672	0.0177	6.85	0.00716
			25th percentile (s)	0.0116	0.0358	0.205	0.669	0.0168	6.81	0.00711
			75th percentile (s)	0.0124	0.0461	0.239	0.681	0.155	6.89	0.00748
Small Batch	–	10	Median (s)	0.0102	0.0475	0.170	0.668	0.0169	4.87	0.00536
			25th percentile (s)	0.00971	0.0333	0.159	0.666	0.0161	4.86	0.00506
			75th percentile (s)	0.0103	0.0540	0.180	0.670	0.0191	4.89	0.00545
Dynamic	–	12	Median (s)	0.0119	0.0419	0.186	0.671	0.0173	6.16	0.00589
			25th percentile (s)	0.0113	0.0377	0.182	0.670	0.0172	6.14	0.00540
			75th percentile (s)	0.0125	0.0630	0.189	0.676	0.0174	6.20	0.00658
		15	Median (s)	0.0123	0.0896	0.296	0.679	0.139	8.04	0.00740
			25th percentile (s)	0.0115	0.0627	0.291	0.676	0.128	8.01	0.00622
			75th percentile (s)	0.0131	1.09	0.321	0.683	0.156	8.09	0.00760
		20	Median (s)	0.0133	1.49	0.367	25.0	0.143	11.2	0.00987
			25th percentile (s)	0.0119	0.447	0.355	24.6	0.130	11.1	0.00927
			75th percentile (s)	0.0154	6.90	0.381	30.1	0.156	11.2	0.00991
		30	Median (s)	0.0149	1.77	7.61	79.7	0.355	17.4	0.0141
			25th percentile (s)	0.0124	1.44	4.88	79.4	0.204	17.2	0.0135
			75th percentile (s)	0.0157	9.24	13.7	80.5	0.364	17.5	0.0145
		37	Median (s)	0.0134	1.77	10.1	81.0	0.376	21.7	0.0173
			25th percentile (s)	0.0118	1.43	5.25	79.7	0.364	21.6	0.0171
			75th percentile (s)	0.0160	9.23	14.2	82.9	0.383	21.8	0.0174
		50	Median (s)	0.0135	1.77	26.3	158	0.732	29.8	0.0233
			25th percentile (s)	0.0129	1.42	17.0	157	0.685	29.7	0.0229
			75th percentile (s)	0.0153	9.21	37.2	160	0.751	30.1	0.0235

N: number of potential reactors of CSTR; B: evaluation budget; –: not applicable; Uniform: uniform sampling without replacement.

Table S13: Grid-GDP: target hits, full-run completion times, summed evaluation costs, and search times of the six DFO methods and the two sampling controls with a budget of 100 evaluations, with 210 runs per method (21 test functions and ten seeds). A target hit is a run that reaches the reference target of Equation (8) of the main paper, a normalized objective within 10^{-5} of the normalized reference objective zero. The summed evaluation cost of a run adds the separately measured costs of all its evaluations, including repeated configurations, the search time is the method time excluding this cost, and the full-run completion time is their sum (Equation (10) of the main paper). Medians and linearly interpolated 25th and 75th percentiles are taken over all runs with recorded times, including target misses, which leaves out the times of the 21 `NOMAD` runs marked a, and separately for each quantity, so the percentiles of the two components need not sum to those of the completion time.

		Full-run completion time (s)			Summed evaluation cost (s)			Search time (s)		
Method	Target hits (of 210)	Median	25th pct.	75th pct.	Median	25th pct.	75th pct.	Median	25th pct.	75th pct.
`D-SDA`	101	18.38	12.42	37.06	18.38	12.41	37.05	0.0024	0.0023	0.0026
`NOMAD`[a]	126	18.50	13.46	35.00	15.20	11.09	30.82	2.04	1.79	2.28
`RBFOpt`	171	70.81	54.83	93.09	60.84	44.31	82.15	7.58	4.12	14.82
`MISO`	181	94.82	79.22	116.05	63.21	44.03	82.37	32.70	25.19	39.38
`BAM`	147	64.42	48.74	81.67	63.97	48.34	81.22	0.43	0.37	0.51
`FMQA-SA`	129	124.13	105.57	139.26	62.34	43.98	77.52	62.04	61.58	62.53
Uniform	45	60.51	43.33	78.60	60.51	43.33	78.59	0.0026	0.0026	0.0027
`BAM`-prefix control	124	61.81	40.96	78.33	61.80	40.95	78.33	0.0026	0.0026	0.0027

pct.: percentile; Uniform: uniform sampling without replacement after the three shared initial configurations; `BAM`-prefix control: the ten predetermined sampling points of `BAM` after the three shared initial configurations, followed by uniform sampling without replacement.
[a] `NOMAD` times cover the 189 runs with recorded times; the 21 runs that crashed on every attempt have none, but their target hits are counted (Section S12).

S8 Data sources and reproducibility

The publication snapshot contains the source code, experimental data, analysis results, and reproduction instructions for every experiment in this article. The `CSTR` updates are the penalty of 10^6 for unaccepted responses, analyzed in `results/cstr_penalty_1e6_analysis_20261022`, and the `CSTR30` evaluation table rebuilt with the mass-balance/zero-recycle initialization, in `results/cstr30_publication_20261025`; the `Column` runs on its 64-configuration logical domain with $B = 13$ are in `results/column_logical_b13_analysis_20261025`. The updated `GDPopt` analysis and execution records contain the 63 Grid and 27 process runs configured with `SCIP` for the mixed-integer master problems and `IPOPT` for the NLP subproblems. The three presolve-disabled LOA diagnostic runs are excluded. The same records contain the paired comparison with the earlier configuration that used `SCIP` for both the mixed-integer master problems and the NLP subproblems, in which GLOA returned the same `CSTR` objectives within a relative difference of 3×10^{-5}. Seed spread, tolerance sensitivity, search overhead, and `GDPopt` enumeration counts are generated by `review_comparison/revision_diagnostics.py` from the retained run records and execution logs. The `D-SDA` rerun, including the process restart variants, is `experiments/dsda_no_revisit/run.py`, which replays the sealed lookup package of the adopted `D-SDA` rerun. The `NOMAD` and `MISO` runs were executed at source revision c4d7db6; the returned records contain the original archive, its SHA-256 checksum, and hash-checked analysis tables of every run. All `D-SDA` runs, including the restart variants, were rerun at source revision c186408, where no configuration is evaluated twice; with re-evaluation switched back on, the same code reproduces all 1,190 published `D-SDA` runs exactly, and the returned records hold every run. The `Lomustine` case is `experiments/lomustine_simulation`, which vendors the corrected-model table of SECQUOIA (2026b) and runs the seven methods through the harness in `experiments/catalytic_distillation_live`; the run records and analysis are in `results/lomustine_simulation_20261031`. The `Catalytic Column` model and its review are in `gdp_models/catalytic_distillation`, the live runs are `experiments/catalytic_distillation_live`, and their returned records, with the original archive, its SHA-256 checksum, and an audit of every run, are in `results/catalytic_distillation_live_20261101`. The MINLP forms, their generators, the fixed-configuration validation, and the solver runs are `experiments/catalytic_distillation_minlp`;

the runs reported here are in `results/catalytic_distillation_minlp_dantzig_20261101`, and an earlier run of the same experiments on a workstation with `GAMS` 54.2.2, in `results/catalytic_distillation_minlp_20261101`, returned the same outcome for every solver, form, and configuration. The `D-SDA` neighborhood comparison of Section S14 is `experiments/dsda_separable`, with results in `results/dsda_separable_20261101`, and the check of the `Catalytic Column` configurations outside the search box is in `results/catalytic_distillation_outside_box_dantzig_20261103`, and its earlier workstation run in `results/catalytic_distillation_outside_box_20261102`. The seven-method tables and figures are generated from these tables and the published records by `review_comparison/seven_methods.py` and the `review_comparison/joint_process_*.py` scripts, without new solver calls.

S9 Feasibility and bound consistency of conventional GDP results

Of the 45 conventional GDP runs in Table S23, 21 have flagged bounds (Table S14) relative to the checked solutions in Table S15. These results combine the 27 new `SCIP`–`IPOPT` `GDPopt` runs with the 18 retained direct `SCIP` runs. Unrounded values, residuals, and source records are linked in Section S8.

For each minimization model, let f_w be the recorded objective of its best independently checked saved feasible solution, and let L denote a reported lower bound. We apply the same rule to all five methods:

$$\tau_w = 10^{-5}\max\{1, |f_w|\}, \qquad \Delta = L - f_w, \qquad \rho = \Delta/\tau_w, \qquad \dagger \iff \rho > 1. \tag{S22}$$

Missing lower bounds are left unreported.

Saved vectors are loaded into the original GDP to reevaluate objectives and check active constraints, variable bounds, integrality, disjunctions, and logical validity without optimization. Violations are computed from the unscaled residuals of the original equations. All 43 saved feasible vectors satisfy the 10^{-5} threshold; the largest violation is 9.90×10^{-7}. The two Big-M runs without accepted feasible solutions remain in the 45-run denominator. This numerical check establishes neither exact feasibility nor global optimality.

Eight of the 21 discrepancies are only 1.09–3.54 times the tolerance and are sensitive to numerical feasibility conventions. These conflicting bounds and `optimal` reports provide no independent global certificate or intrinsic method ranking; their cause in `GDPopt` or its subsolvers remains unresolved.

Table S14: Process GDP models: the 21 conventional GDP runs whose reported lower bound exceeds the recorded objective of the comparison solution of the same model by more than the audit tolerance, from one run per method and model. The comparison solution of each model is the checked feasible solution of Table S15, with recorded objective f_w, and each row gives the reported lower bound of the run, the largest violation at its saved solution, and the excess of the bound over f_w in multiples of the audit tolerance (Section S9). Displayed values are rounded, and the flags use unrounded values.

Model	N	Method	ID	Reported L	Unit	$v_{\max}$	C	ρ
`CSTR`	5	GLOA	S02	3.13020	L	9.85×10^{-12}	C01	$2.23 \times 10^{3\dagger}$
		LOA	S03	3.06201	L	2.60×10^{-11}	C01	$1.29^{\dagger}$
		RIC	S01	3.06201	L	2.60×10^{-11}	C01	$1.29^{\dagger}$
		Hull	S05	3.06201	L	9.34×10^{-7}	C01	$1.09^{\dagger}$
	10	GLOA	S07	3.44796	L	3.11×10^{-15}	C02	$1.93 \times 10^{4\dagger}$
		LOA	S08	2.96605	L	1.62×10^{-10}	C02	$2.65 \times 10^{3\dagger}$
		RIC	S06	2.96605	L	1.62×10^{-10}	C02	$2.65 \times 10^{3\dagger}$
		Hull	S10	2.88951	L	9.37×10^{-7}	C02	$3.54^{\dagger}$
	15	GLOA	S12	3.56196	L	1.01×10^{-15}	C03	$2.59 \times 10^{4\dagger}$
		LOA	S13	2.82939	L	1.33×10^{-15}	C03	$1.30^{\dagger}$
		RIC	S11	2.82939	L	1.33×10^{-15}	C03	$1.30^{\dagger}$
	20	GLOA	S17	3.61612	L	4.60×10^{-11}	C04	$2.92 \times 10^{4\dagger}$
		LOA	S18	2.79941	L	1.58×10^{-15}	C04	$1.67^{\dagger}$
		RIC	S16	2.79941	L	1.58×10^{-15}	C04	$1.67^{\dagger}$
	25	GLOA	S22	3.64786	L	2.22×10^{-14}	C05	$3.12 \times 10^{4\dagger}$
		RIC	S21	2.79125	L	1.38×10^{-10}	C05	$3.53 \times 10^{2\dagger}$
	30	GLOA	S27	3.66872	L	7.11×10^{-15}	C06	$3.25 \times 10^{4\dagger}$
`Column`	–	GLOA	S32	20,355.1	none	1.15×10^{-13}	C07	$4.65 \times 10^{3\dagger}$
		Hull	S35	20,044.2	none	1.00×10^{-8}	C07	$3.06 \times 10^{3\dagger}$
`Dynamic`	–	GLOA	S42	-197.140	none	9.85×10^{-12}	C09	$7.01 \times 10^{3\dagger}$
		Hull	S45	-196.931	none	1.00×10^{-7}	C09	$7.11 \times 10^{3\dagger}$

f_w: recorded objective of the comparison solution; N: number of potential reactors of `CSTR`; –: not applicable; GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; all three are `GDPopt` methods; Hull: the reformulation solved directly by `SCIP`. ID: run identifier; Reported L: lower bound reported by the run; Unit: unit of the objective, as in Table 1 of the main paper, with none for no physical unit; C: identifier of the comparison solution in Table S15. $v_{\max}$: largest unscaled violation of the original GDP constraints and variable bounds at the saved solution of the run; $\rho = (L - f_w)/\tau_w$, with the audit tolerance $\tau_w = 10^{-5} \max\{1, |f_w|\}$; †: $\rho > 1$.

Table S15: Process GDP models: comparison solutions of Table S14, one for each model, from the conventional GDP runs with one run per method and model. The comparison solution of a model is the saved solution with the lowest recorded objective f_w among the conventional GDP runs whose saved solutions pass the independent feasibility check in the original GDP at 10^{-5}, and the method is the conventional GDP method that returned it. Displayed values are rounded.

Model	N	Method	ID	Recorded f_w	Unit	$v_{\max}$
`CSTR`	5	Big-M	C01	3.06198	L	9.40×10^{-7}
	10	Big-M	C02	2.88941	L	9.76×10^{-7}
	15	Hull	C03	2.82936	L	9.47×10^{-7}
	20	Hull	C04	2.79936	L	9.90×10^{-7}
	25	Hull	C05	2.78143	L	9.90×10^{-7}
	30	Big-M	C06	2.76957	L	9.68×10^{-9}
`Column`	–	RIC	C07	19,449.9	none	1.01×10^{-13}
`Small Batch`	–	Big-M	C08	167,428	$	9.00×10^{-7}
`Dynamic`	–	Big-M	C09	-212.009	none	1.00×10^{-7}

f_w: recorded objective of the comparison solution; N: number of potential reactors of `CSTR`; –: not applicable; RIC: relaxation with integer cuts of `GDPopt`; Big-M, Hull: reformulations solved directly by `SCIP`. ID: identifier used in column C of Table S14; Unit: unit of the objective, as in Table 1 of the main paper, with none for no physical unit; $v_{\max}$: largest unscaled violation of the original GDP constraints and variable bounds at the solution.

Additional tightened-feasibility check for `CSTR15`. An `IPOPT` solve kept the discrete configuration and starting values of an archived LOA solution and targeted an original-GDP residual of 10^{-8} (SECQUOIA, 2026a). That solution differs from the comparison solution C03 of Table S15. Only exactly satisfied constant rows were omitted from the computational clone; the independent check retained all original constraints. The returned objective was 2.82939, with maximum unscaled violation 1.11×10^{-16}. The new GLOA lower bound exceeds this objective by 0.732565, or approximately 2.59×10^4 times the audit tolerance. The discrepancy persists under the tighter check, but its cause remains unresolved.

Additional formulation check for `Dynamic` Hull. The saved Big-M point has objective -212.009, approximately 15.0776 below the Hull lower bound -196.931. It satisfies both the original GDP and a reconstructed Hull formulation at 10^{-5}, with maximum violation approximately 10^{-7}. The cause of the discrepancy remains unresolved.

S10 Additional candidate-search and sampling comparisons

Figure S1 shows the evaluation at which each process run that reaches the target first reaches it, for `CSTR30`, `Column`, `Small Batch`, and `Dynamic` at its largest budget.

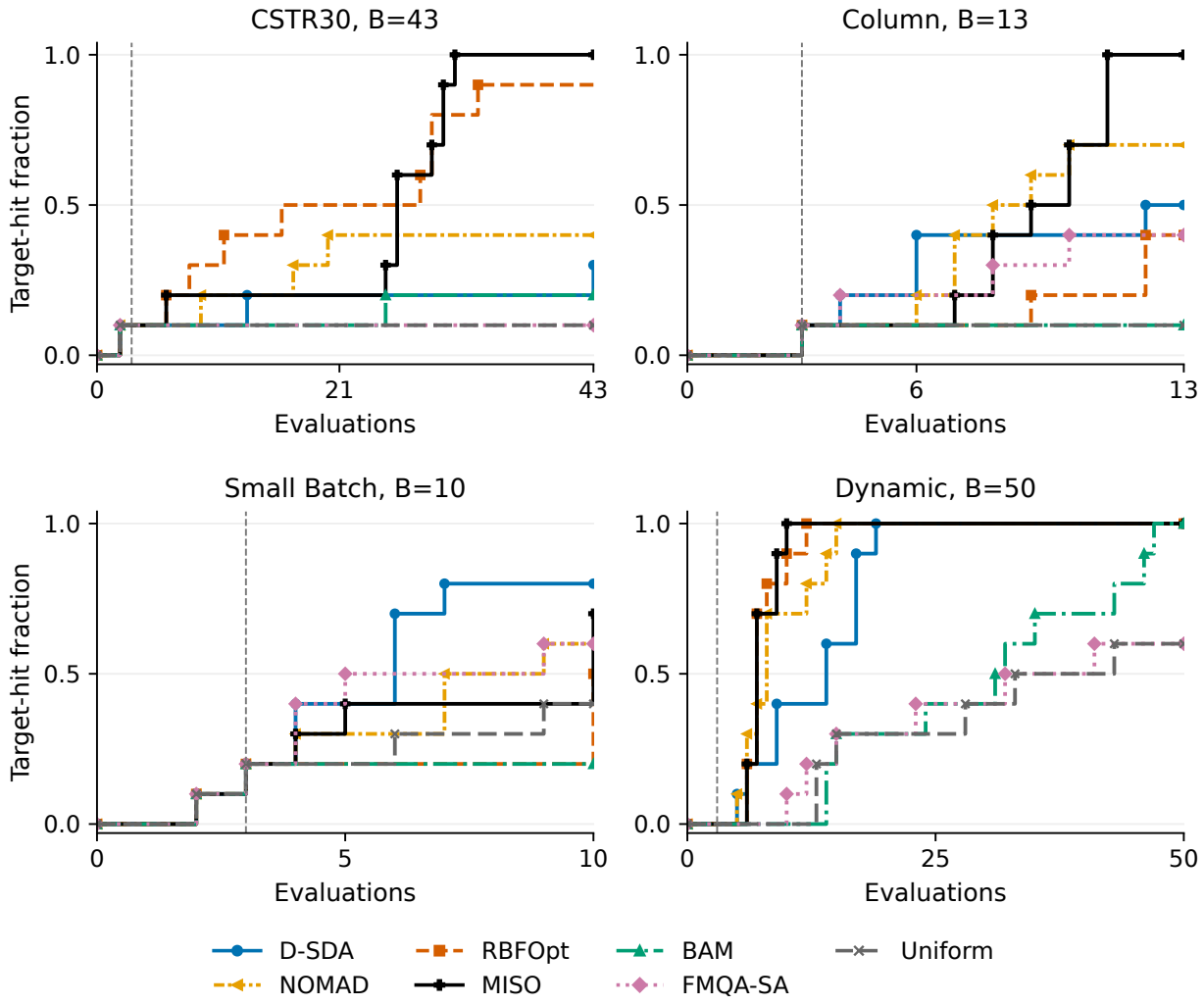


Figure S1: Process GDP models: fraction of the ten runs per method that have reached the reference target of Equation (8) of the main paper, an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1 of the main paper, within a given number of evaluations, for four conditions and the six DFO methods and uniform sampling. Panels: `CSTR30` at $B = 43$, `Column` at $B = 13$, `Small Batch` at $B = 10$, and `Dynamic` at $B = 50$, where B is the evaluation budget. For each method and seed, a `Dynamic` run with $B = 12$, 15, 20, 30, or 37 reaches the target at the same evaluation as the run with $B = 50$ when that evaluation falls within its budget and misses it otherwise, so the `Dynamic` panel also gives the profiles of the smaller budgets up to their budgets. Every evaluation counts, including the three shared initial configurations, unaccepted evaluations, and repeated configurations, and the dashed vertical line at three evaluations marks the end of the shared initial evaluations. Each curve steps up at the evaluation in which a run first observes an accepted objective that meets the target; runs that miss the target remain in the denominator. Methods are distinguished by color, line style, and marker. Uniform: uniform sampling without replacement. Target hits in all 14 process conditions are listed in Table 2 of the main paper.

S10.1 Paired differences and sampling controls

The paired comparisons quantify differences in target-hit rates under shared initial designs, matching case, budget, and seed. Positive differences favor the first method and are expressed in percentage points. For Grid, the descriptive percentile bootstrap resamples the 21 intact function vectors, each containing ten paired seed differences, because the ten seeds of a function share its landscape and are not independent draws. For Process, paired seeds are resampled within each of the 14 conditions; the six related `Dynamic` budgets are not pooled for inference. The seed is 20261020 with 2,000 bootstrap replications and deterministic case/seed ordering. Intervals are exploratory and unadjusted for multiple comparisons; ten-seed boundary cases may yield degenerate intervals, and zero inclusion does not establish equivalence.

An interval entirely above zero favors the first method under this resampling; an interval spanning zero leaves the direction uncertain. There is no preset cutoff for practical importance. For example, `BAM` matches its prefix control at $B = 10$ but exceeds it by 10.95 percentage points at $B = 100$, so initial sampling alone does not account for its final target-hit rate.

Table S16: Grid-GDP and process GDP models: seed-paired differences in the target-hit percentage of selected pairs of methods and their 95% percentile bootstrap intervals, with 210 paired runs per Grid-GDP budget (21 functions and ten seeds) and ten per process condition. The difference is the percentage of runs of the first method that reach the reference target of Equation (8) of the main paper minus that of the second, so positive values favor the first method. The intervals come from 2,000 bootstrap replications that resample the 21 functions for Grid-GDP and the ten seeds within the stated condition only for the process models, and they are not adjusted for multiple comparisons. For the process conditions, the seed outcomes count the seeds in which both methods, only the first, only the second, or neither reaches the target.

				95% interval (pp)		Seed outcomes			
Benchmark	B	Contrast	Difference (pp)	Lower	Upper	Both	First only	Second only	Neither
Grid-GDP	10	`D-SDA` − `RBFOpt`	-13.81	-24.29	-6.19				
	100	`D-SDA` − `RBFOpt`	-33.33	-48.10	-19.52				
	10	`RBFOpt` − Uniform	13.33	5.71	24.29				
	100	`RBFOpt` − Uniform	60.00	42.86	75.71				
	10	`BAM` − `BAM`-prefix control	0.00	0.00	0.00				
	100	`BAM` − `BAM`-prefix control	10.95	0.48	23.81				
	10	`FMQA-SA` − Uniform	0.95	-0.95	2.86				
	100	`FMQA-SA` − Uniform	40.00	28.57	52.38				
`Small Batch`	10	`FMQA-SA` − Uniform	20	-30	60	2	4	2	2
`Dynamic`	12	`RBFOpt` − `D-SDA`	60	30	90	4	6	0	0

B: evaluation budget; pp: percentage points; Contrast: first method − second method; Uniform: uniform sampling without replacement; `BAM`-prefix control: the ten predetermined sampling points of `BAM` after the three shared initial configurations, followed by uniform sampling without replacement.

S10.2 Exact paired tests for process conditions

Table S17 lists the seed-paired target outcomes of `RBFOpt` and `D-SDA` in every process condition and of `BAM` and `RBFOpt` on `CSTR5`. The p-values come from the two-sided exact McNemar test, which uses only the pairs in which exactly one method reaches the target. They are descriptive and not adjusted for multiple comparisons. In the 60 paired `CSTR` runs, `RBFOpt` reaches the target in 30 of the 31 discordant pairs.

Table S17: Process GDP models: seed-paired target outcomes of `RBFOpt` and `D-SDA` and of `BAM` and `RBFOpt`, with ten seed pairs per condition. The counts give the seeds in which both methods, only the first, only the second, or neither reaches the reference target of Equation (8) of the main paper. The p-value uses only the seeds in which exactly one method reaches the target, and the p-values are descriptive and not adjusted for multiple comparisons. The pooled `CSTR` row combines the six sizes, with 60 seed pairs.

Model	N	B	Both	First only	Second only	Neither	p
First `RBFOpt`, second `D-SDA`							
`CSTR`	5	8	2	6	0	2	0.0312
	10	15	2	4	1	3	0.375
	15	22	2	5	0	3	0.0625
	20	29	3	4	0	3	0.125
	25	36	3	5	0	2	0.0625
	30	43	3	6	0	1	0.0312
	all	–	15	30	1	14	2.98×10^{-8}
`Column`	–	13	3	1	2	4	1
`Small Batch`	–	10	4	1	4	1	0.375
`Dynamic`	–	12	4	6	0	0	0.0312
		15	6	4	0	0	0.125
		20	10	0	0	0	1
		30	10	0	0	0	1
		37	10	0	0	0	1
		50	10	0	0	0	1
First `BAM`, second `RBFOpt`							
`CSTR`	5	8	8	2	0	0	0.500

N: number of potential reactors of `CSTR`; B: evaluation budget; all: the six sizes pooled; –: not applicable in the N column and several budgets in the B column; p: two-sided exact McNemar p-value.

S10.3 Grid suite exclusions

Excluding Xin-She Yang 3 and Bohachevsky 2 preserves the main descriptive pattern. For the sampled Bohachevsky 2 lattice, $\cos(3\pi a_i) = 1$ at every coordinate, leaving an objective nearly redundant with sampled Sum of Squares after normalization. Without both functions, $B = 100$ rates are 84.7% for `MISO`, 79.5% for `RBFOpt`, 66.8% for `BAM`, 57.9% for `FMQA-SA`, 55.8% for `NOMAD`, 54.7% for the `BAM`-prefix control, 42.6% for `D-SDA`, and 17.9% for uniform. At $B = 10$, `BAM` and its prefix control each retain 47.9%, compared with 15.8% for `RBFOpt`, 14.7% for `MISO`, and 12.6% for `NOMAD`.

S11 Seed spread, tolerance sensitivity, and search overhead

Table S18 compares exact target attainment with relaxed relative tolerances for the 980 primary process runs. The relative gap is $(f_{\text{best}} - f_{\text{ref}})/\max\{1, |f_{\text{ref}}|\}$, the normalization of the target rule in Equation (8) of the main paper. The exact target-hit rows apply the target tolerance of 10^{-5}. Distinct finals is the largest number of different final objectives among the ten seeds of any condition in the family, and the smallest and largest miss-gap rows cover every run that misses the exact target. Table S19 compares the median configuration-evaluation cost with the median search time per evaluation. Both are medians over runs of per-run averages, namely the summed point cost or the stored search time divided by the number of evaluations, over all seven methods for the evaluation cost. The 21 `NOMAD` Grid runs without a recorded search time are left out of its search median.

Every NLP subproblem solved by `GDPopt` in the process comparisons fixes a different configuration, as recorded in its execution logs before each subproblem. RIC and LOA solve one subproblem for each of the $N(N+1)/2$ logically admissible configurations of `CSTR5` to `CSTR20` and for each of the 37 admissible `Dynamic` configurations, and RIC also covers all 325 admissible configurations of `CSTR25`.

Table S18: Process GDP models: exact and relaxed target attainment and the spread of final objectives of the six DFO methods and uniform sampling, with ten runs per method and condition, pooled over the six sizes of `CSTR` and the six budgets of `Dynamic`. An exact target hit is a run that reaches the reference target of Equation (8) of the main paper, with relative tolerance 10^{-5}, and the relaxed rows count the runs whose relative gap $(f_{\text{best}} - f_{\text{ref}})/\max\{1, |f_{\text{ref}}|\}$ is at most the stated value. The miss gaps are the smallest and largest relative gaps over the runs that miss the exact target, and the distinct-finals row gives the largest number of different final objectives among the ten seeds of any condition of the model.

Model		D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
	Exact target hits (of 60)	16	21	45	43	18	9	11
	Runs with gap $\leq 0.1\%$	16	21	45	43	18	9	11
	Runs with gap $\leq 1\%$	19	21	45	43	60	23	25
`CSTR`	Runs with gap $\leq 5\%$	27	27	47	49	60	41	48
	Distinct finals	5	3	3	5	3	10	10
	Smallest miss gap (%)	0.224	2.23	2.23	2.23	0.224	0.169	0.224
	Largest miss gap (%)	13.0	13.0	12.5	10.6	0.692	17.2	14.3
	Exact target hits (of 10)	5	7	4	10	1	4	1
	Runs with gap $\leq 0.1\%$	5	7	4	10	1	4	1
	Runs with gap $\leq 1\%$	8	9	5	10	1	4	2
`Column`	Runs with gap $\leq 5\%$	10	10	9	10	10	10	9
	Distinct finals	4	3	6	1	2	4	8
	Smallest miss gap (%)	0.536	0.536	0.536	–	1.05	1.05	0.536
	Largest miss gap (%)	1.14	1.14	5.92	–	1.05	2.76	5.22
	Exact target hits (of 10)	8	6	5	7	2	6	4
	Runs with gap $\leq 0.1\%$	8	6	5	7	2	6	4
	Runs with gap $\leq 1\%$	8	6	5	7	2	6	4
`Small Batch`	Runs with gap $\leq 5\%$	8	6	5	7	2	6	4
	Distinct finals	2	2	3	2	3	5	4
	Smallest miss gap (%)	22.2	6.64	6.64	6.64	6.64	6.64	6.64
	Largest miss gap (%)	22.2	6.64	22.2	6.64	8.23	22.2	22.2
	Exact target hits (of 60)	50	58	60	60	27	23	21
	Runs with gap $\leq 0.1\%$	50	58	60	60	27	23	21
	Runs with gap $\leq 1\%$	50	58	60	60	27	23	21
`Dynamic`	Runs with gap $\leq 5\%$	54	60	60	60	46	42	38
	Distinct finals	5	3	1	1	4	7	6
	Smallest miss gap (%)	2.07	2.07	–	–	2.07	2.07	2.07
	Largest miss gap (%)	36.3	2.66	–	–	7.01	36.3	30.7

f_{best}: best accepted objective of the run; f_{ref}: reference objective of Table 1 of the main paper; Uniform: uniform sampling without replacement; –: no run misses the exact target.

Table S19: Grid-GDP and process GDP models: median evaluation cost and median search time per evaluation of the six DFO methods and uniform sampling, with 1,050 Grid-GDP runs per method (21 functions, five budgets, and ten seeds) and ten runs per method and process condition. The evaluation cost of a run is the summed recorded cost of its evaluated configurations divided by its number of evaluations, and its median pools the runs of all seven methods. The search time per evaluation of a run is its search time, the method time excluding the evaluation cost, divided by its number of evaluations. Medians are over runs, and the 21 `NOMAD` Grid-GDP runs without a recorded search time are left out of the `NOMAD` median.

Benchmark	Median evaluation cost (s)	Median search time per evaluation (ms)						
		D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
Grid-GDP	0.390	0.0777	50.5	31.1	231	3.77	604	0.0336
`CSTR`	1.46	1.24	72.2	24.2	287	9.01	572	0.517
`Column`	2.50	0.941	3.25	17.4	51.7	1.36	527	0.551
`Small Batch`	0.422	1.02	4.75	17.0	66.8	1.69	487	0.536
`Dynamic`	0.836	0.893	96.9	98.4	1,860	9.34	567	0.471

Uniform: uniform sampling without replacement.

S12 NOMAD and MISO runs

The NOMAD and MISO runs use the stored responses, shared initial configurations, seeds, budgets, penalties, and reference targets of the other five methods. Neither method receives the target as a stopping criterion. They were executed after the other methods, on a shared server with six workers, each restricted to one logical CPU on a different physical core and to one OpenMP and BLAS thread; the cores were not reserved exclusively. The server used NumPy 2.4.6 and SciPy 1.17.1; the other versions are given in Section 3.6 of the main paper. As for the other methods, completion times add the measured search time of each run to the recorded cost of every evaluated configuration, so they do not depend on the speed of the server's NLP solves.

Termination. All 1,190 MISO runs use their full budget. NOMAD stops before the budget in 77 of the 140 process runs and in 697 of the 1,050 Grid runs, including all 210 runs at $B = 100$, where it stops after a median of 31.5 evaluations. Its Grid target fraction is therefore 60.0% at both $B = 70$ and $B = 100$.

NOMAD crashes and reruns. Of the 1,190 NOMAD runs, 1,046 completed on the first attempt. The other 144 ended with a segmentation fault after their last new evaluation. Of these, 110 completed on the first rerun and 13 on the second, each after evaluating the same configurations in the same order as the crashed attempt, and 21 crashed on all three attempts after identical evaluations. All 21 are Grid runs at $B = 100$, and 16 of them had reached the target before the crash. Scoring them on the evaluations completed before the crash gives NOMAD 526 of 1,050 Grid targets and 92 of 140 process targets; counting all 21 as misses gives 510 Grid targets, or 52.4% instead of 60.0% at $B = 100$. These 21 runs have no recorded search time, so the NOMAD timing summaries at Grid $B = 100$ use the other 189 runs. For a completed rerun, the search time is that of the completed attempt; the time spent in crashed attempts is not recorded. No MISO run failed.

Exploratory paired tests. Table S20 compares NOMAD and MISO with D-SDA, RBFOpt, and each other in every process condition, a family of 70 exact McNemar tests defined after the runs were completed. With a Holm correction over the 70 tests, the smallest adjusted p-value is 0.547, for MISO against NOMAD on CSTR20, so no contrast is significant at the 0.05 level. Ten seeds give each test little power, and a nonsignificant test does not establish equivalence. The six Dynamic budgets reuse the same seeds and are not independent replications. The last row pools the six CSTR sizes for the descriptive comparisons in the main text; it is not part of the corrected family.

Table S20: Process GDP models: seed-paired target outcomes of NOMAD and MISO against D-SDA, RBFOpt, and each other, with ten seed pairs per condition. The discordant seeds are those in which only one method of the column reaches the reference target of Equation (8) of the main paper, and the p-values are not adjusted for multiple comparisons. With a Holm correction over the 70 per-condition tests, the smallest adjusted p-value is 0.547. The pooled CSTR row combines the six sizes, with 60 seed pairs, and is not part of the 70 per-condition tests.

Model	N	B		NOMAD vs D-SDA	NOMAD vs RBFOpt	MISO vs D-SDA	MISO vs RBFOpt	MISO vs NOMAD
CSTR	5	8	Discordant seeds	2:0	1:5	2:0	0:4	1:1
			p	0.500	0.219	0.500	0.125	1
	10	15	Discordant seeds	0:0	1:4	0:1	0:4	0:1
			p	1	0.375	1	0.125	1
	15	22	Discordant seeds	0:0	0:5	5:0	1:1	5:0
			p	1	0.0625	0.0625	1	0.0625
	20	29	Discordant seeds	0:1	0:5	7:0	3:0	8:0
			p	1	0.0625	0.0156	0.250	0.00781
	25	36	Discordant seeds	4:1	2:4	7:0	2:0	4:0
			p	0.375	0.688	0.0156	0.500	0.125
	30	43	Discordant seeds	1:0	1:6	7:0	1:0	6:0
			p	1	0.125	0.0156	1	0.0312
Column	–	13	Discordant seeds	3:1	4:1	5:0	6:0	3:0
			p	0.625	0.375	0.0625	0.0312	0.250
Small Batch	–	10	Discordant seeds	1:3	2:1	1:2	2:0	3:2
			p	0.625	1	1	0.500	1
Dynamic	–	12	Discordant seeds	4:0	0:2	6:0	0:0	2:0
			p	0.125	0.500	0.0312	1	0.500
		15	Discordant seeds	4:0	0:0	4:0	0:0	0:0
			p	0.125	1	0.125	1	1
		20	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		30	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		37	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		50	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
CSTR	all	–	Discordant seeds	7:2	5:29	28:1	7:9	24:2
			p	0.180	3.86×10^{-5}	1.12×10^{-7}	0.804	1.05×10^{-5}

a:b: discordant seeds in which only the first or only the second method of the column reaches the target; p: unadjusted two-sided exact McNemar p-value; N: number of potential reactors of CSTR; B: evaluation budget; all: the six sizes pooled; –: not applicable in the N column and several budgets in the B column.

S13 Process D-SDA restarts

The process D-SDA of the main comparison descends once from the best of the three shared initial configurations (S). Variant R applies the Grid rule and descends from all three shared initial configurations, best first, while budget remains. Variant RR continues after R with descents from uniformly drawn unvisited configurations until the budget is exhausted, and it uses the full budget in all 140 runs. No variant receives the reference target. All variants read the stored responses and, like the main comparison, never evaluate a configuration twice. Table S21 shows that R and RR evaluate 1,055 and 1,249 more configurations than S and add one target, on seed 4 of CSTR30, so target attainment rises from 79 to 80 of the 140 runs. Both restart variants improve the final objective in the same 14 CSTR runs. In each of them, S ends at $(5, 1)$, 8.3 to 13.0% above the reference, and the restarts end at the target in the one new hit and 0.4 to 8.6% above it in the other 13. On CSTR10 to CSTR30, 31 of the 36 single-descent misses end at $(5, 1)$.

Table S21: Process GDP models: target hits, full-budget runs, median evaluations, and median full-run completion times of three D-SDA restart variants, with ten runs per variant and condition. A target hit is a run that reaches the reference target of Equation (8) of the main paper, and a full-budget run uses all B evaluations, as every RR run does. The completion time of a run adds its measured search time to the recorded cost of every evaluated configuration, and medians are over the ten runs.

			Target hits (of 10)			Full budget (of 10)		Median evaluations			Median completion time (s)		
Model	N	B	S	R	RR	S	R	S	R	RR	S	R	RR
CSTR	5	8	2	2	2	8	10	8	8	8	10.67	10.71	10.71
	10	15	3	3	3	3	10	11.5	15	15	17.61	21.57	21.57
	15	22	2	2	2	2	10	15.5	22	22	25.68	35.19	35.19
	20	29	3	3	3	0	10	19	29	29	32.31	47.49	47.49
	25	36	3	3	3	2	9	23.5	36	36	45.49	60.78	60.78
	30	43	3	4	4	1	8	24.5	43	43	47.17	78.88	78.88
Column	–	13	5	5	5	9	10	13	13	13	35.44	35.69	35.69
Small Batch	–	10	8	8	8	10	10	10	10	10	4.23	4.23	4.23
Dynamic	–	12	4	4	4	9	10	12	12	12	12.22	12.25	12.25
		15	6	6	6	6	10	15	15	15	15.32	15.76	15.76
		20	10	10	10	3	10	18	20	20	20.02	20.76	20.76
		30	10	10	10	0	10	18	30	30	20.02	27.13	27.13
		37	10	10	10	0	2	18	34.5	37	20.02	28.01	30.05
		50	10	10	10	0	0	18	34.5	50	20.02	28.01	37.00

S: single descent from the best shared initial configuration; R: descents from the three shared initial configurations, best first, while budget remains; RR: R followed by descents from uniformly drawn unvisited configurations until the budget is exhausted. N: number of potential reactors of CSTR; B: evaluation budget; –: not applicable.

S14 `D-SDA` neighborhood

The process, Grid, and `Lomustine D-SDA` runs use the unit L_∞ neighborhood, whose $3^d - 1$ neighbors number 8 for $d = 2$ external variables and 26 for $d = 3$. The `Catalytic Column`, with $d = 6$ and 728 such neighbors, uses the separable neighborhood, which changes one external variable by ± 1 and has at most $2d$ neighbors. To test the effect of this choice, we repeated every `D-SDA` run on stored responses with the separable neighborhood and all other settings unchanged: the 1,050 Grid runs, the 140 single-descent process runs, and the 180 `Lomustine` runs in the six option orders of Section S5. Table S22 pairs each run with its L_∞ counterpart.

The separable neighborhood reaches more targets when the budget is small compared with the L_∞ neighborhood: on Grid at $B = 10$ (22 against 13 of 210 runs, with 9 and 0 discordant runs, exact McNemar $p = 0.004$), on `Dynamic` at $B = 12$ and 15 (16 against 10 of 20, $p = 0.031$), and on `Lomustine` over the six orders (99 against 90 of 180, $p = 0.022$). Its descents also stop earlier: from $B = 50$, the Grid runs end after a median of 25 evaluations in place of 37, and they reach 88 targets against 101 at $B = 70$ and 100 ($p = 0.011$). On `CSTR`, the separable neighborhood reaches 7 of the 60 targets against 16 ($p = 0.004$). Eighteen of its 53 misses stop on the edge $\xi_1 = \xi_2$ of the admissible region, where every one-coordinate move leaves the region or worsens the objective, whereas a diagonal move continues along the edge to the target at (N, N). Over the 140 process runs, the two neighborhoods reach 79 and 78 targets. Neither neighborhood dominates: the L_∞ neighborhood suits two external variables with budgets that cover its eight neighbors, and the separable neighborhood suits more external variables or smaller budgets.

Table S22: Grid-GDP, process GDP models, and `Lomustine`: target hits and median evaluations of `D-SDA` with the unit L_∞ neighborhood and with the separable neighborhood, with 210 runs of each neighborhood per Grid-GDP row (21 functions and ten seeds), ten per process row, and 60 per `Lomustine` row (six option orders and ten seeds). Both neighborhoods run on stored responses with all other settings unchanged, the process runs use the single descent, and each separable run is paired with its L_∞ counterpart. A target hit is a run that reaches the reference target of Equation (8) of the main paper, and the median evaluations are medians of the number of evaluations per run.

			Target hits				Median evaluations	
Benchmark	N	B	L_∞	Separable	Separable only	L_∞ only	L_∞	Separable
Grid-GDP	–	10	13	**22**	9	0	10	10
		30	78	**79**	14	13	30	25
		50	**91**	87	10	14	37	25
		70	**101**	88	5	18	37	25
		100	**101**	88	5	18	37	25
`CSTR`	5	8	2	2	0	0	8	8
	10	15	**3**	1	0	2	11.5	8
	15	22	**2**	1	0	1	15.5	10.5
	20	29	**3**	1	0	2	19	14
	25	36	**3**	1	0	2	23.5	15.5
	30	43	**3**	1	0	2	24.5	17
`Column`	–	13	5	**6**	3	2	13	12
`Small Batch`	–	10	8	**9**	2	1	10	10
`Dynamic`	–	12	4	**6**	2	0	12	12
		15	6	**10**	4	0	15	13
		20	10	10	0	0	18	13
		30	10	10	0	0	18	13
		37	10	10	0	0	18	13
		50	10	10	0	0	18	13
`Lomustine`	–	10	23	**27**	6	2	10	10
		15	32	**36**	4	0	15	11
		20	35	**36**	1	0	19	11

L_∞: unit neighborhood, which changes every external variable by at most one; Separable: neighborhood that changes one external variable by ± 1; Separable only, L_∞ only: paired runs in which only that neighborhood reaches the target. N: number of potential reactors of `CSTR`; B: evaluation budget; –: not applicable. Bold: larger number of target hits of the two neighborhoods in each row; equal counts are not bold.

S15 Joint comparison with conventional GDP methods

Figures S2 and S3 compare the eight process conditions not shown in the main joint completion-time figure. Each DFO condition has ten seeds, whereas each conventional GDP method has one run per problem. Timing and target criteria follow Figure 3 of the main paper. Table S23 reports all nine models, including final objectives and termination flags.

Table S23: Process GDP models: final objectives and full-run completion times of the conventional GDP methods, with one run per method and model. The `GDPopt` methods GLOA, LOA, and RIC use `SCIP` for the master problems and `IPOPT` for the NLP subproblems, and the Big-M and Hull columns are the retained runs that solve these reformulations directly with `SCIP`. The final objective is in the unit of the row label, as in Table 1 of the main paper, with no physical unit for `Column` and `Dynamic`, and the completion time is the measured full-run time, including model-template preparation. The success threshold of a conventional GDP run is a final objective of at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$, and feasibility means numerical feasibility under the original-GDP residual test at 10^{-5}, which does not establish exact feasibility. Runs that do not stop at the time limit report `optimal` termination.

Model	N		GLOA	LOA	RIC	Big-M	Hull
`CSTR`	5	Final objective (L)	3.13020†‡	3.06201†	3.06201†	**3.06198**	3.06201†
		Completion time (s)	**4.57**	24.58	15.38	53.68	15.43
	10	Final objective (L)	3.44796†‡	2.96605†‡	2.96605†‡	**2.88941**	2.88951†
		Completion time (s)	**3.10**	140.38	134.31	771.36	71.63
	15	Final objective (L)	3.56196†‡	2.82939†	2.82939†	–^{TL}	**2.82936**
		Completion time (s)	**6.17**	445.43	283.77	1800+	284.63
	20	Final objective (L)	3.61612†‡	2.79941†	2.79941†	2.80413^{TL}	**2.79936**
		Completion time (s)	**7.46**	733.53	607.72	1800+	1,150.67
	25	Final objective (L)	3.64786†‡	2.79125^{TL}	2.79125†‡	2.78771^{TL}	**2.78143**
		Completion time (s)	**5.20**	1800+	1,042.79	1800+	1,701.75
	30	Final objective (L)	3.66872†‡	2.87314^{TL}	2.77162^{TL}	**2.76957**^{TL}	**2.76957**^{TL}
		Completion time (s)	**5.52**	1800+	1800+	1800+	1800+
`Column`	–	Final objective	20,355.1†‡	**19,449.9**	**19,449.9**	–^{TL}	22,271.5$^{\text{TL}\dagger}$
		Completion time (s)	**42.41**	199.22	124.03	1800+	1800+
`Small Batch`	–	Final objective ($)	**167,428**	**167,428**	**167,428**	**167,428**	**167,428**
		Completion time (s)	10.32	5.61	13.15	0.51	**0.50**
`Dynamic`	–	Final objective	-197.140†‡	**-212.009**	**-212.009**	**-212.009**^{TL}	-196.931†‡
		Completion time (s)	238.96	550.49	**149.39**	1800+	921.28

N: number of potential reactors of `CSTR`; f_{ref}: reference objective of Table 1 of the main paper; GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts. 1800+: stopped at the 1,800 s time limit; TL: stopped at the time limit, with reported `maxTimeLimit`; –: no accepted feasible solution in an objective cell and not applicable in the N column. †: reported lower bound exceeds the best checked saved numerically feasible objective f of the model by more than $10^{-5}\max\{1, |f|\}$ (Section S9); ‡: final objective above the success threshold, despite reported `optimal` termination and a better checked numerically feasible solution. Bold: lowest final objective of each model, including ties within $10^{-5}\max\{1, |f_{\min}|\}$ of that lowest objective $f_{\min}$, and shortest completion time of each model, including ties.

Table S24: Process GDP models: NLP subproblems solved per run by the six DFO methods and uniform sampling, with ten runs per method and condition, and by the single run of each `GDPopt` method, which uses `SCIP` for the master problems and `IPOPT` for the NLP subproblems. The DFO entries are medians over the ten runs of the number of evaluations that called `IPOPT`, whether or not its result was accepted, without logical and preprocessing rejections, and the `GDPopt` entries are the `IPOPT` calls confirmed in the execution log of the run. The counts are numbers of subproblem solves, which differ in cost, and do not include solver iterations; the internal NLP counts of the direct Big-M and Hull solves are unavailable and are not treated as zero. Every `GDPopt` NLP call fixes a different configuration, out of $N(N+1)/2$ logically admissible configurations for `CSTR` with N potential reactors and 37 for `Dynamic` (Section S11). The `GDPopt` methods have no evaluation budget, so the single `Dynamic` run of each appears in every `Dynamic` row.

			DFO methods and uniform sampling (median of ten)							`GDPopt` (one run)		
Model	N	B	`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform	GLOA	LOA	RIC
`CSTR`	5	8	6	6	6	6	6	5.5	4.5	4	15	15
	10	15	10.5	10.5	12	11.5	8.5	9	7.5	2	55	55
	15	22	14	15	16.5	16.5	11	15	11	4	120	120
	20	29	18	20	20.5	19.5	17	17.5	15.5	4	210	210
	25	36	22	25	25.5	25	18.5	23	21	2	315	325
	30	43	23	25	31.5	29.5	21.5	30.5	21.5	2	243	455
`Column`	–	13	11	10.5	12	11	7.5	7	7	18	38	38
`Small Batch`	–	10	5	6	6	6	5	5	4	12	5	12
`Dynamic`	–	12	10	10	10	9	8	6.5	6	20	37	37
		15	12.5	12	13	12	9	8	6.5	20	37	37
		20	16.5	14.5	16	14	11	11	9.5	20	37	37
		30	16.5	14.5	21	21	17	14.5	15	20	37	37
		37	16.5	14.5	25	25	19	19	19	20	37	37
		50	16.5	14.5	31	29	26	24.5	26	20	37	37

N: number of potential reactors of `CSTR`; B: evaluation budget of the DFO methods; –: not applicable; Uniform: uniform sampling without replacement. GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts.

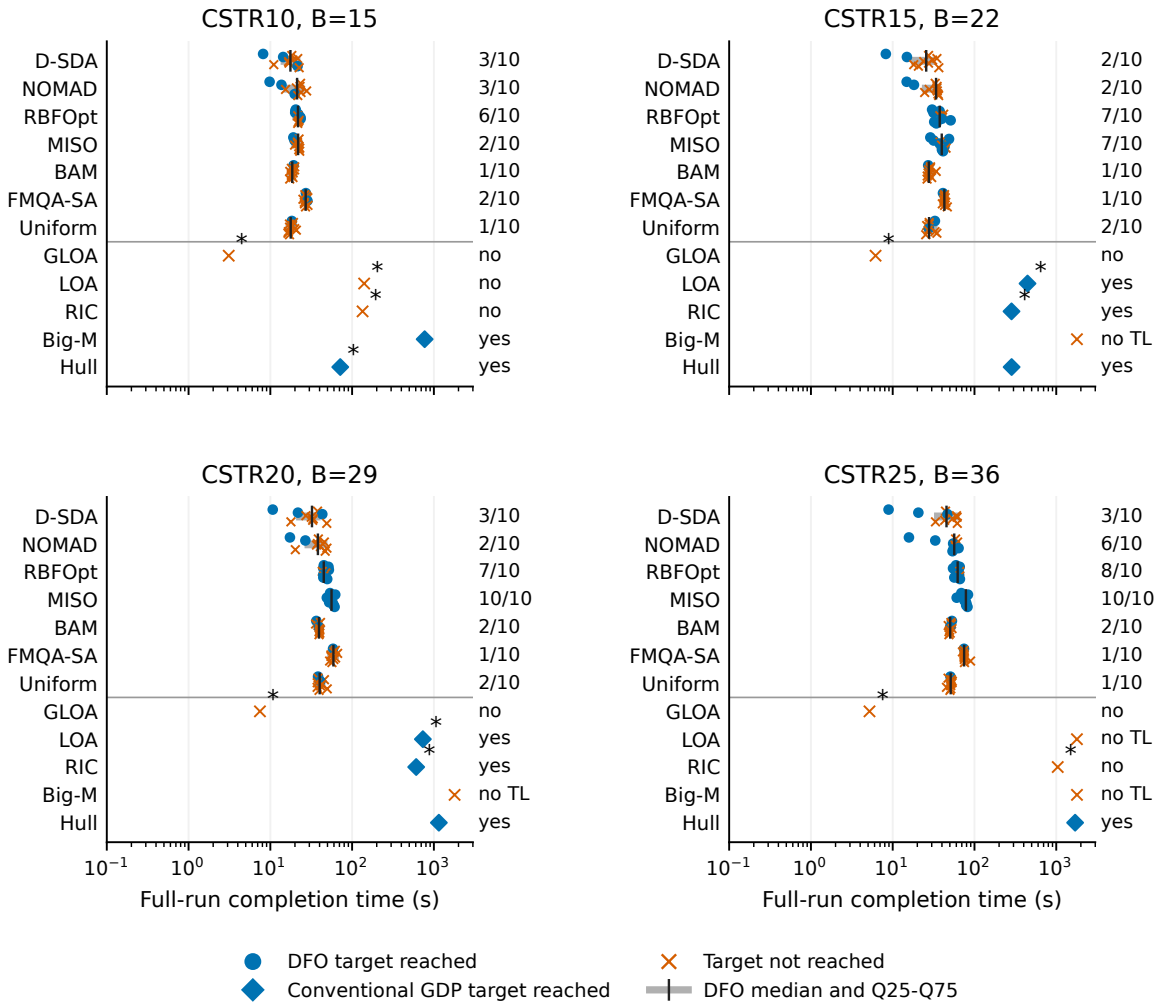


Figure S2: Process GDP models: full-run completion times on `CSTR10` to `CSTR25`, which Figure 3 of the main paper does not show, of the six DFO methods and uniform sampling (ten runs each, at the evaluation budget B of each panel title) and of the single run of each conventional GDP method. GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; Big-M and Hull: reformulations solved by `SCIP`. DFO times add the search time to the measured cost of each evaluated configuration (Equation (10) of the main paper); conventional GDP times are measured and include model preparation. Dots mark DFO runs that reach the reference target of Equation (8) of the main paper, an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1 of the main paper; diamonds mark conventional GDP runs whose feasible objective is at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$; crosses mark the other runs; gray bars span the 25th to 75th percentiles of the ten DFO runs, with a tick at the median. The right-hand labels give the target hits out of ten of each DFO method and the outcome, yes or no, of each conventional run, with TL for runs that report termination at the 1,800 s time limit. Asterisks mark conventional runs whose reported lower bound exceeds a checked feasible objective (Section S9). Uniform: uniform sampling without replacement.

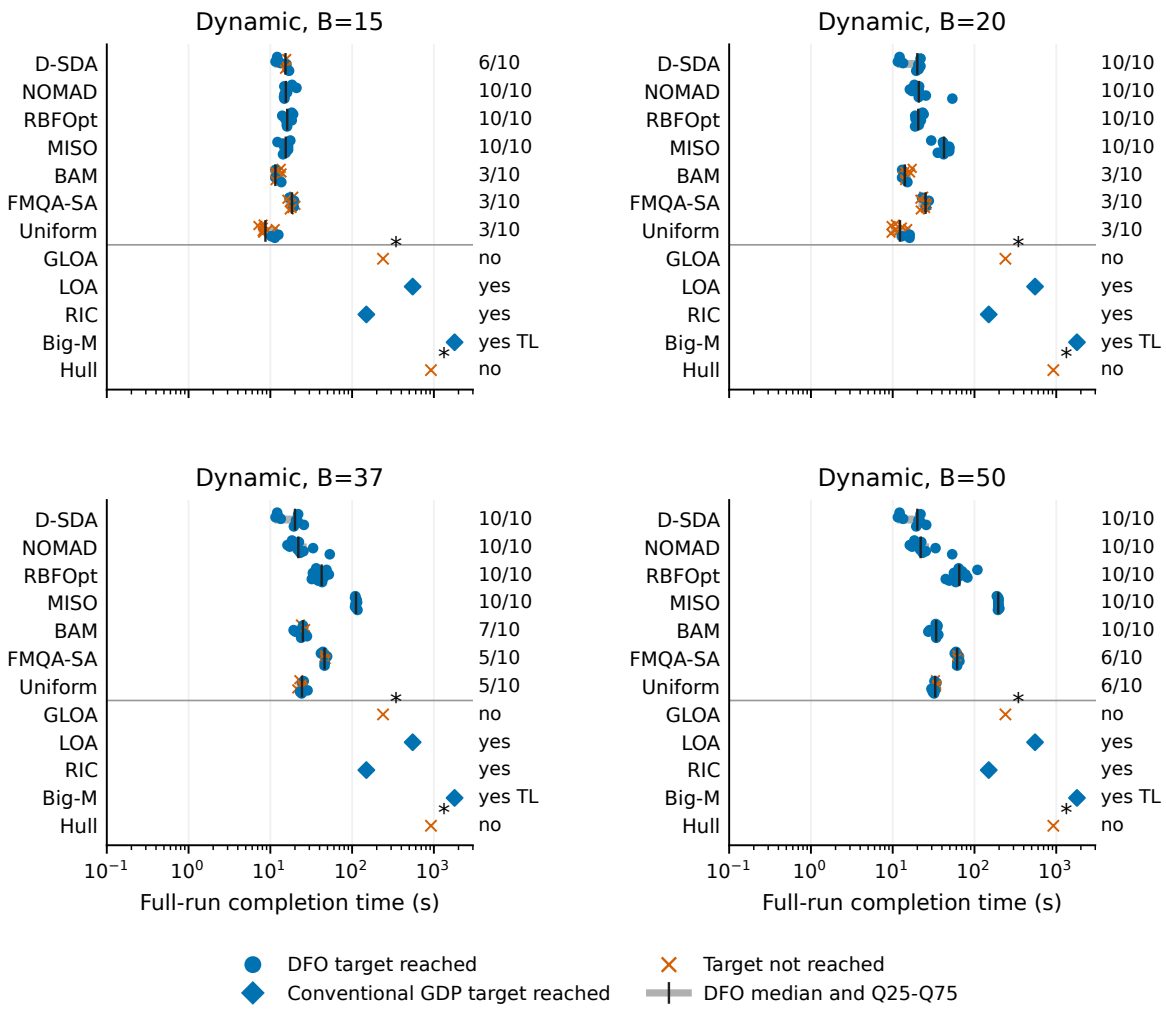


Figure S3: Process GDP models: full-run completion times on `Dynamic` at $B = 15$, 20, 37, and 50, which Figure 3 of the main paper does not show, of the six DFO methods and uniform sampling (ten runs each, at the evaluation budget B of each panel title) and of the single run of each conventional GDP method. GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; Big-M and Hull: reformulations solved by `SCIP`. DFO times add the search time to the measured cost of each evaluated configuration (Equation (10) of the main paper); conventional GDP times are measured and include model preparation. Dots mark DFO runs that reach the reference target of Equation (8) of the main paper, an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1 of the main paper; diamonds mark conventional GDP runs whose feasible objective is at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$; crosses mark the other runs; gray bars span the 25th to 75th percentiles of the ten DFO runs, with a tick at the median. The right-hand labels give the target hits out of ten of each DFO method and the outcome, yes or no, of each conventional run, with TL for runs that report termination at the 1,800 s time limit. Asterisks mark conventional runs whose reported lower bound exceeds a checked feasible objective (Section S9). The conventional GDP methods have no evaluation budget, so the four panels show the same conventional runs. Uniform: uniform sampling without replacement.

References

D-Wave Systems, n.d. dwave-samplers: Classical algorithms for solving binary quadratic models. URL: https://github.com/dwavesystems/dwave-samplers/releases/tag/1.6.0. Version 1.6.0; classical CPU simulated annealing; accessed October 15, 2026.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization. International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/1412.6980.

SECQUOIA, 2026a. Black-Box-for-GDP: benchmark code, experimental data, and analysis. GitHub. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/de6ab9395e5121f27db3b95844b1cd381264367d. commit de6ab939; accessed 2 November 2026.

SECQUOIA, 2026b. pd_ising: Lomustine superstructure PharmaPy simulation workflow and data. URL: https://github.com/SECQUOIA/pd_ising/tree/633cf923f07abba88970fa1fa816c4d06769751e/ds-mfg. pull request 28; commit 633cf923f07abba88970fa1fa816c4d06769751e; accessed October 31, 2026.

Wu Group, n.d. Quantum Hamiltonian descent: 2D test functions. https://jiaqileng.github.io/quantum-hamiltonian-descent/nonconvex2d.html. Source collection for the 21 adapted finite grids used here; accessed September 29, 2026.